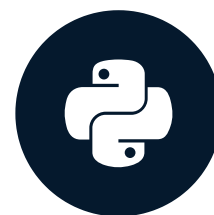


To tool or not to tool!

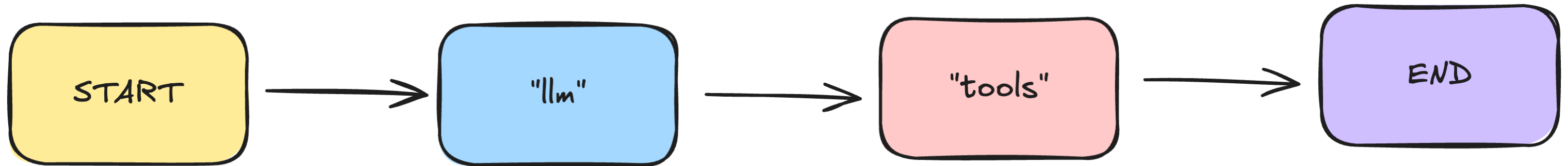
MULTI-AGENT SYSTEMS WITH LANGGRAPH



James Chapman

Curriculum Manager, DataCamp

The linear agent

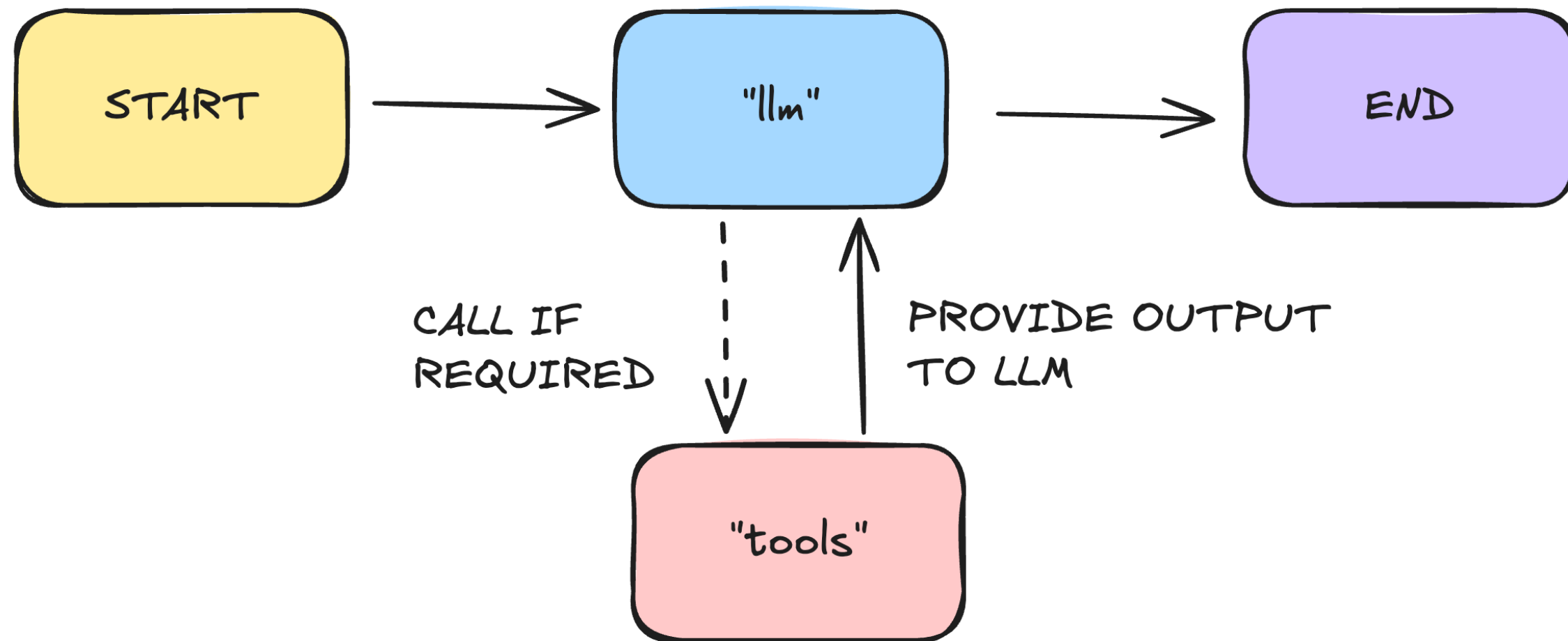


Could respond to some queries

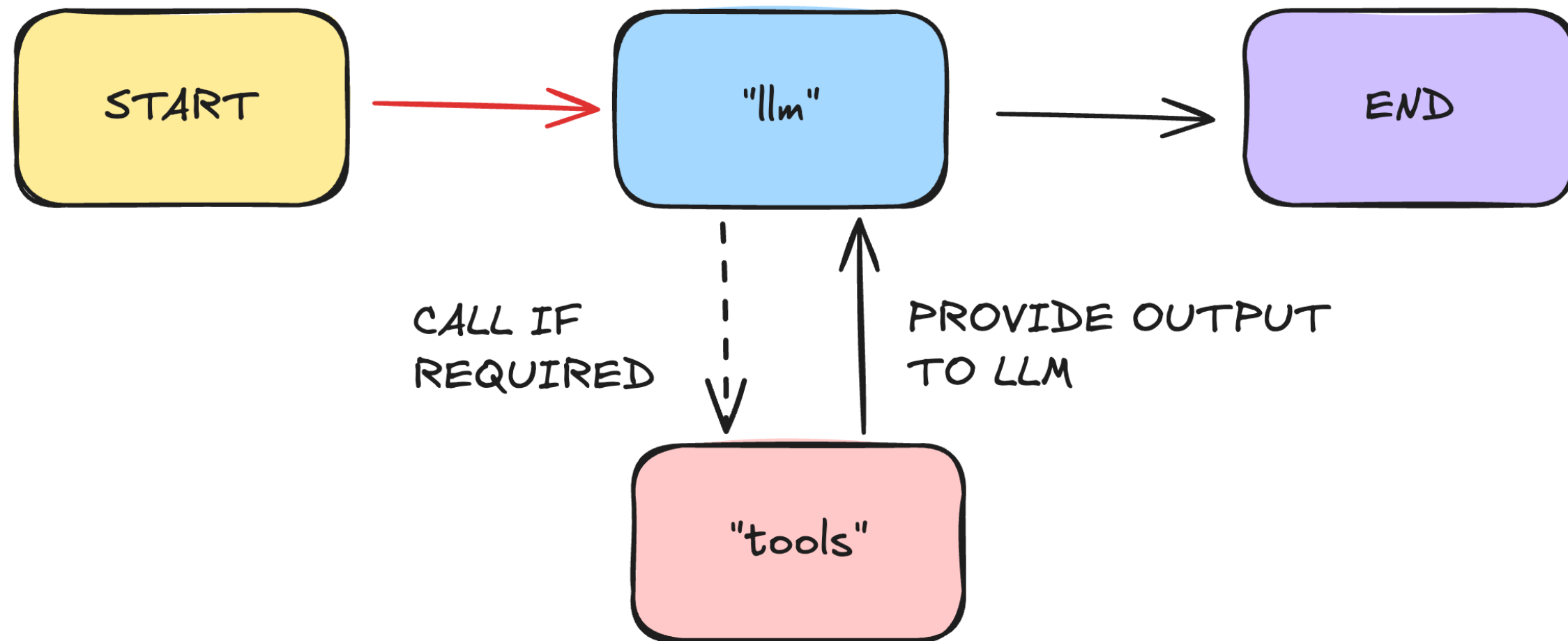
Couldn't handle queries that don't require tools

Couldn't generate plots (requires *recursion*)

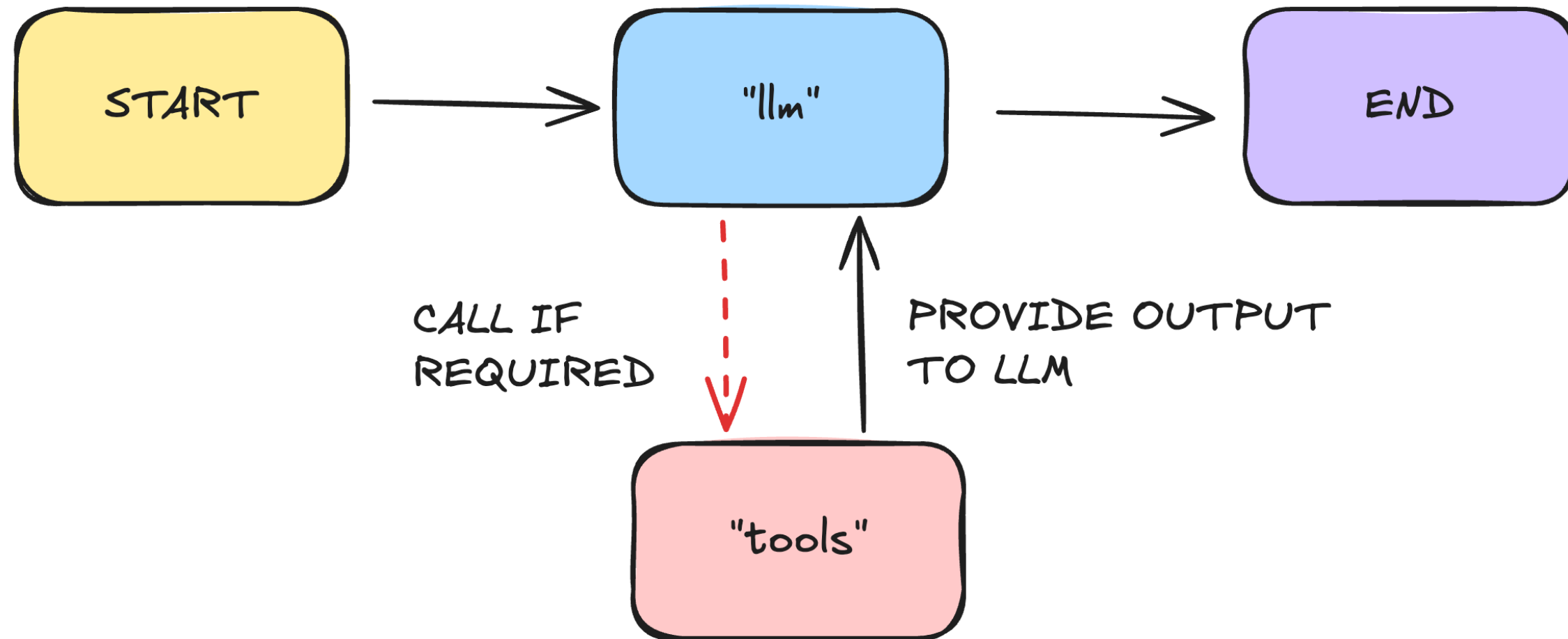
Conditional edges



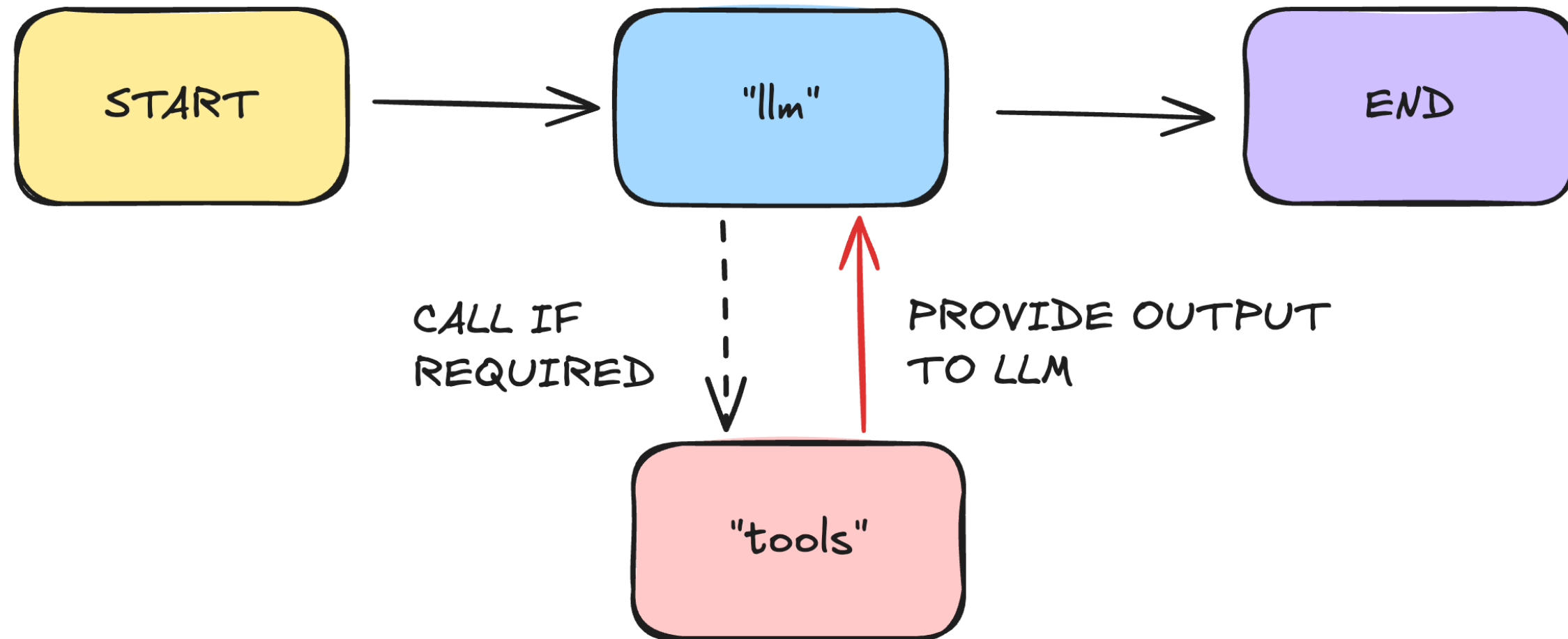
Conditional edges



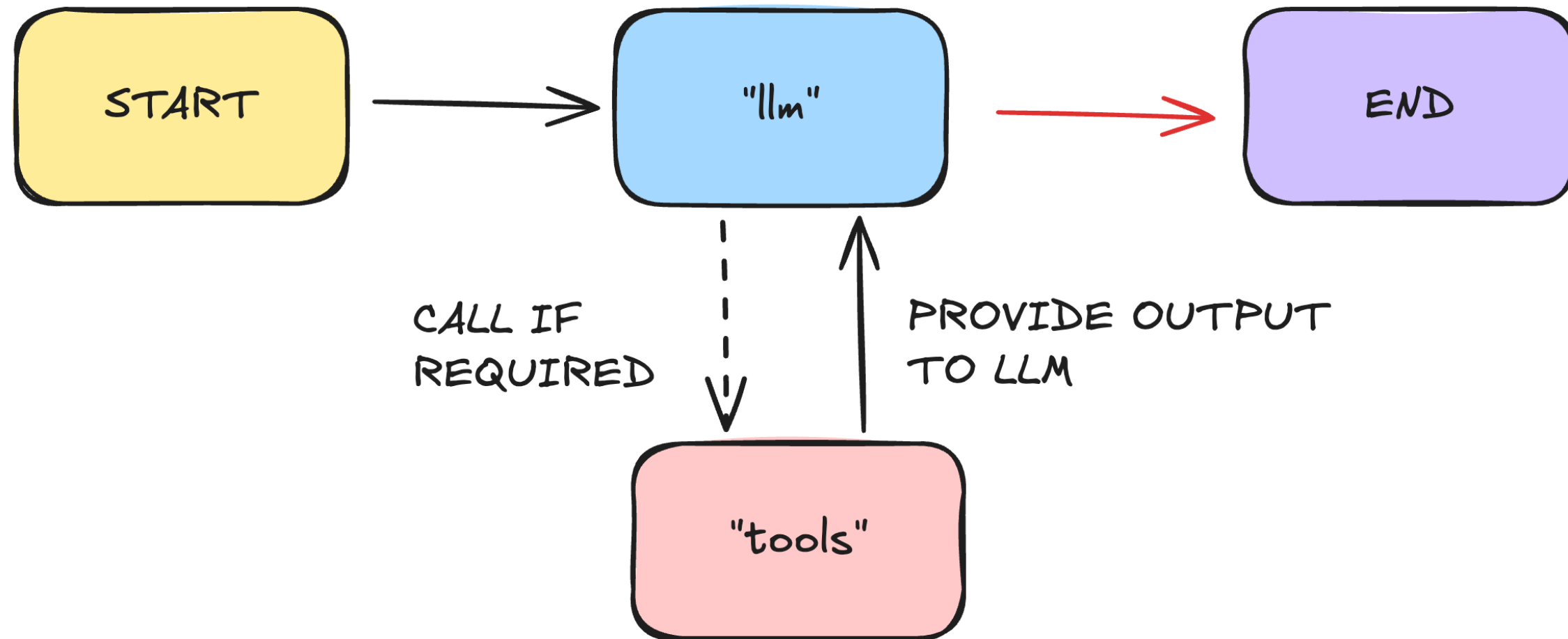
Conditional edges



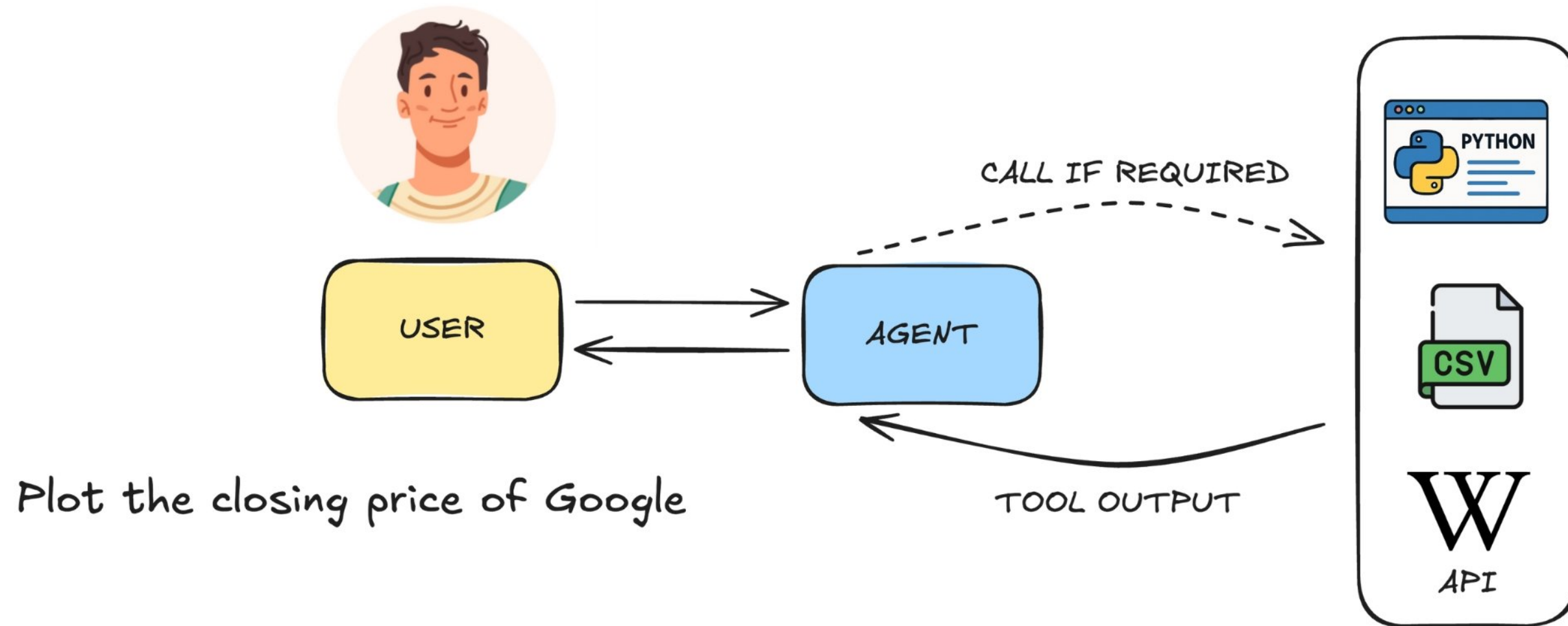
Conditional edges



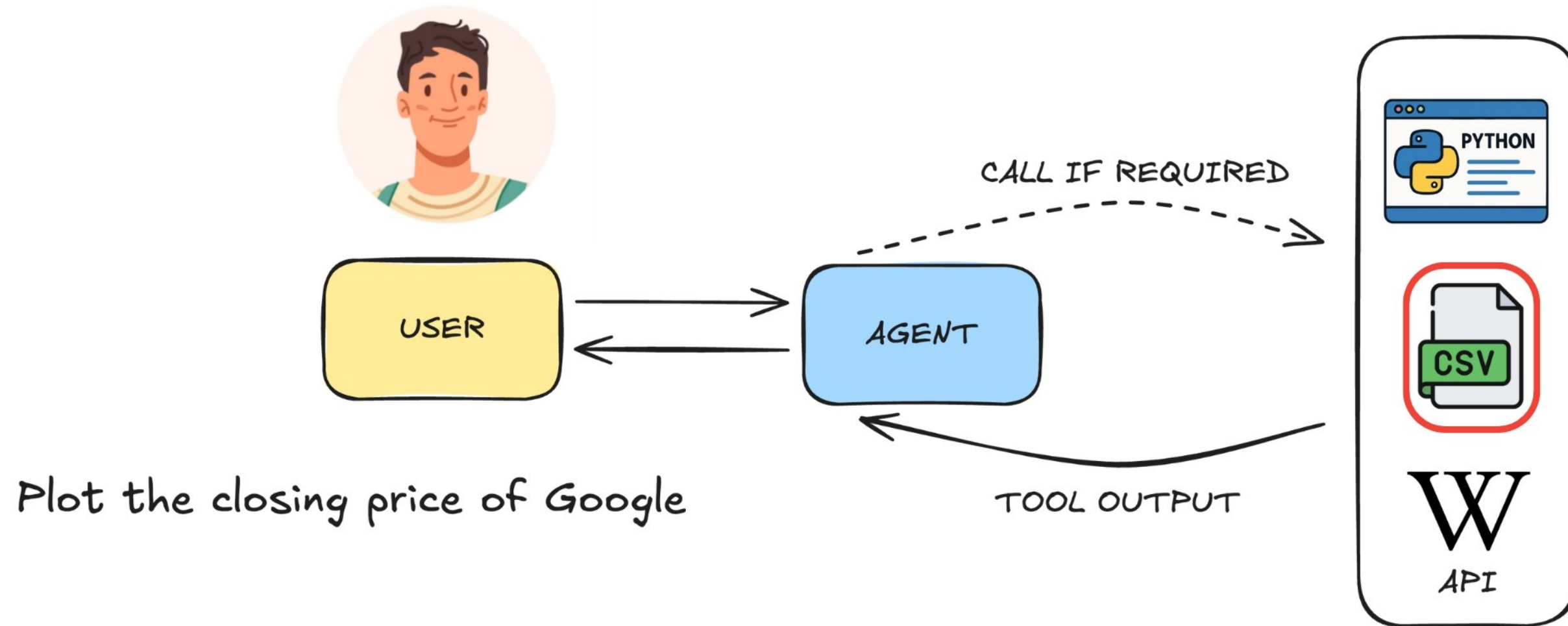
Conditional edges



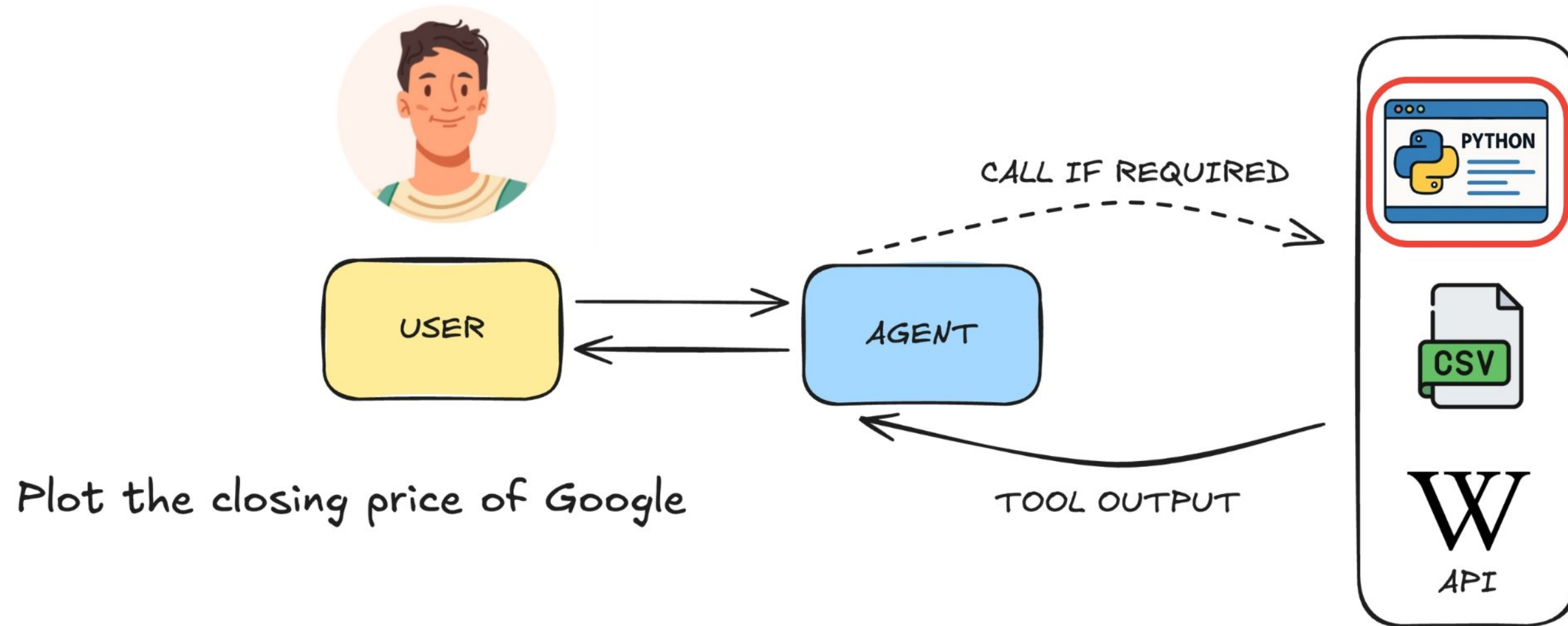
Conditional edges in practice



Conditional edges in practice



Conditional edges in practice

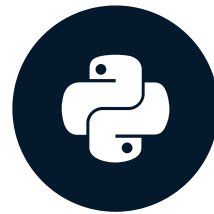


Let's practice!

MULTI-AGENT SYSTEMS WITH LANGGRAPH

Conditional tool calls

MULTI-AGENT SYSTEMS WITH LANGGRAPH



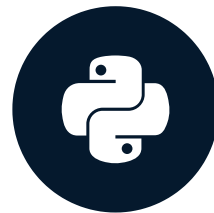
Full Name
Instructor

Let's practice!

MULTI-AGENT SYSTEMS WITH LANGGRAPH

One-line agents

MULTI-AGENT SYSTEMS WITH LANGGRAPH



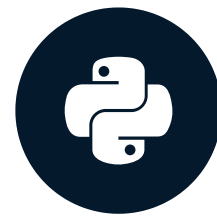
Full Name
Instructor

Let's practice!

MULTI-AGENT SYSTEMS WITH LANGGRAPH

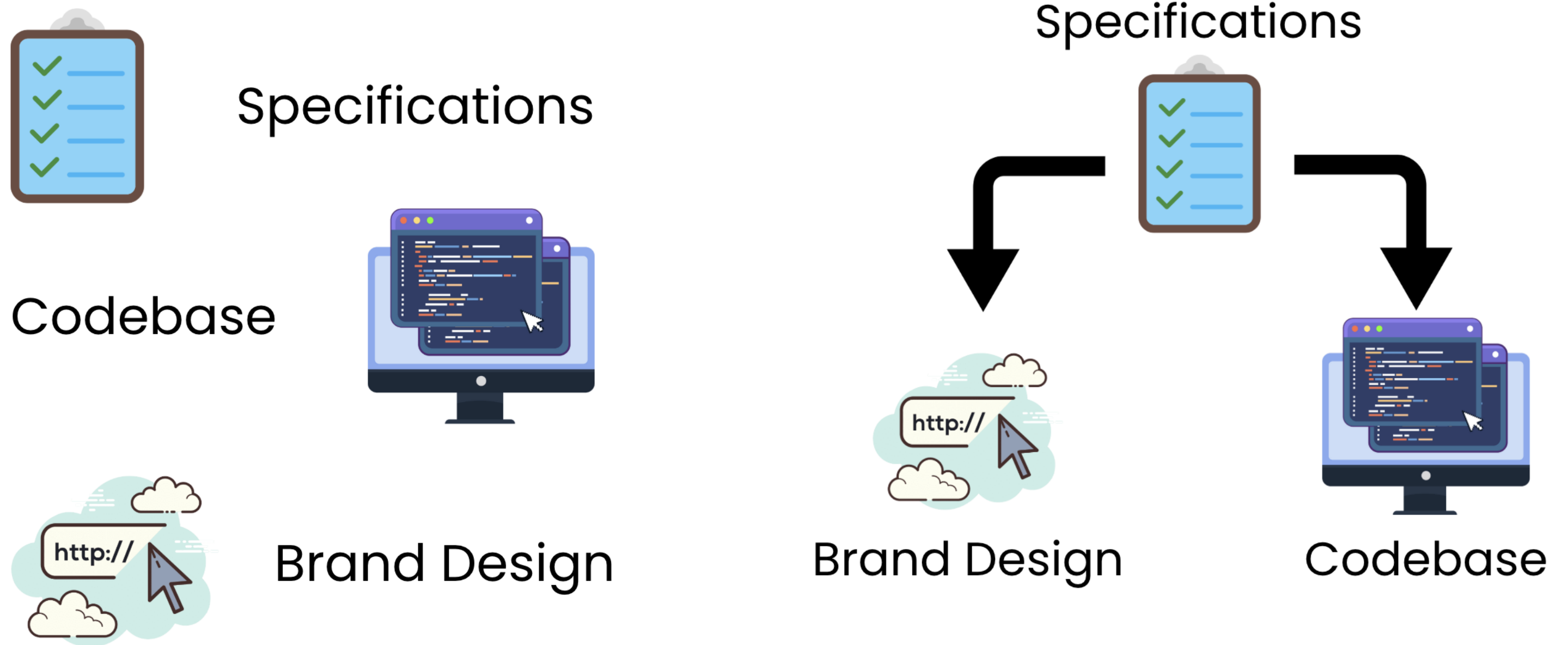
Multi-agent frameworks

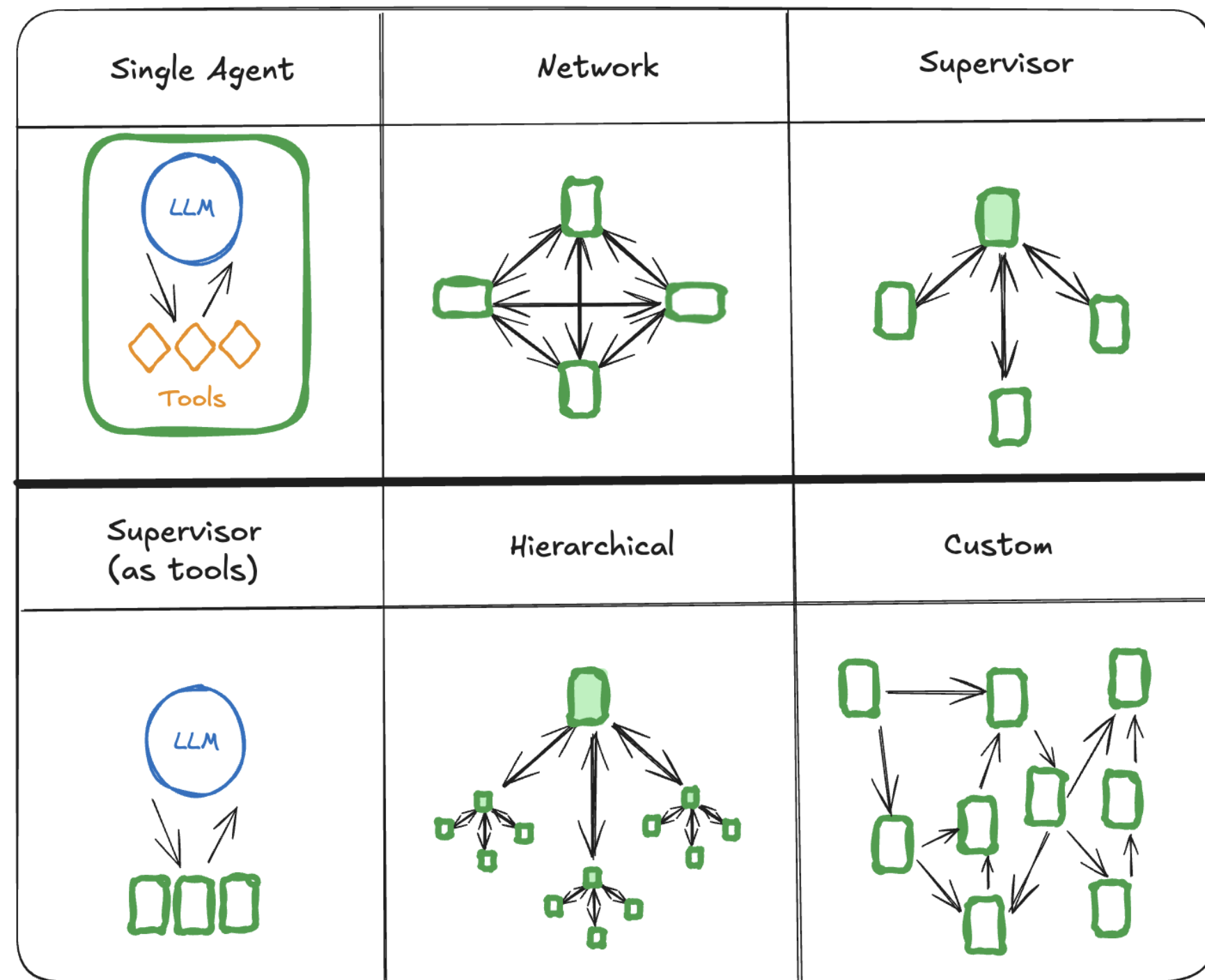
MULTI-AGENT SYSTEMS WITH LANGGRAPH



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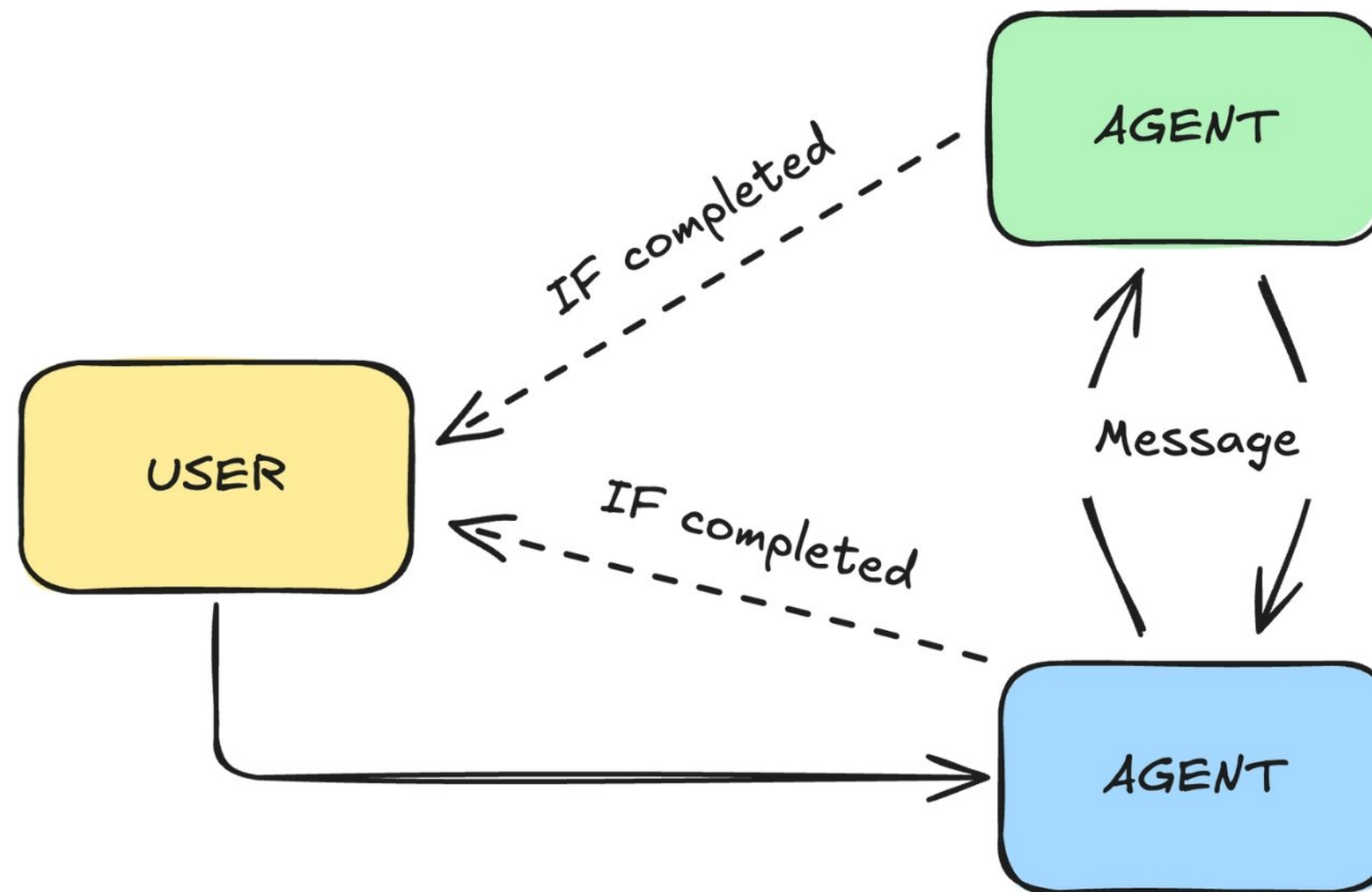
Example: website creator system



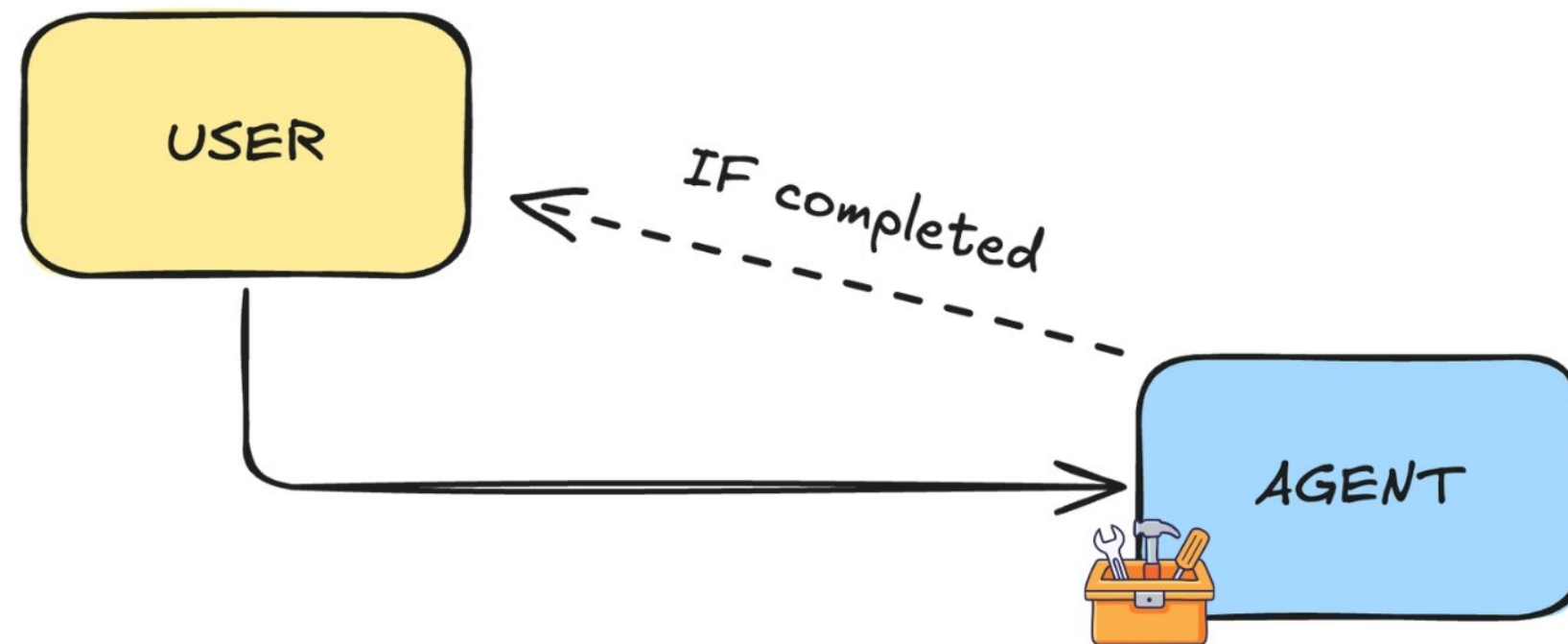


¹ Credit: LangGraph

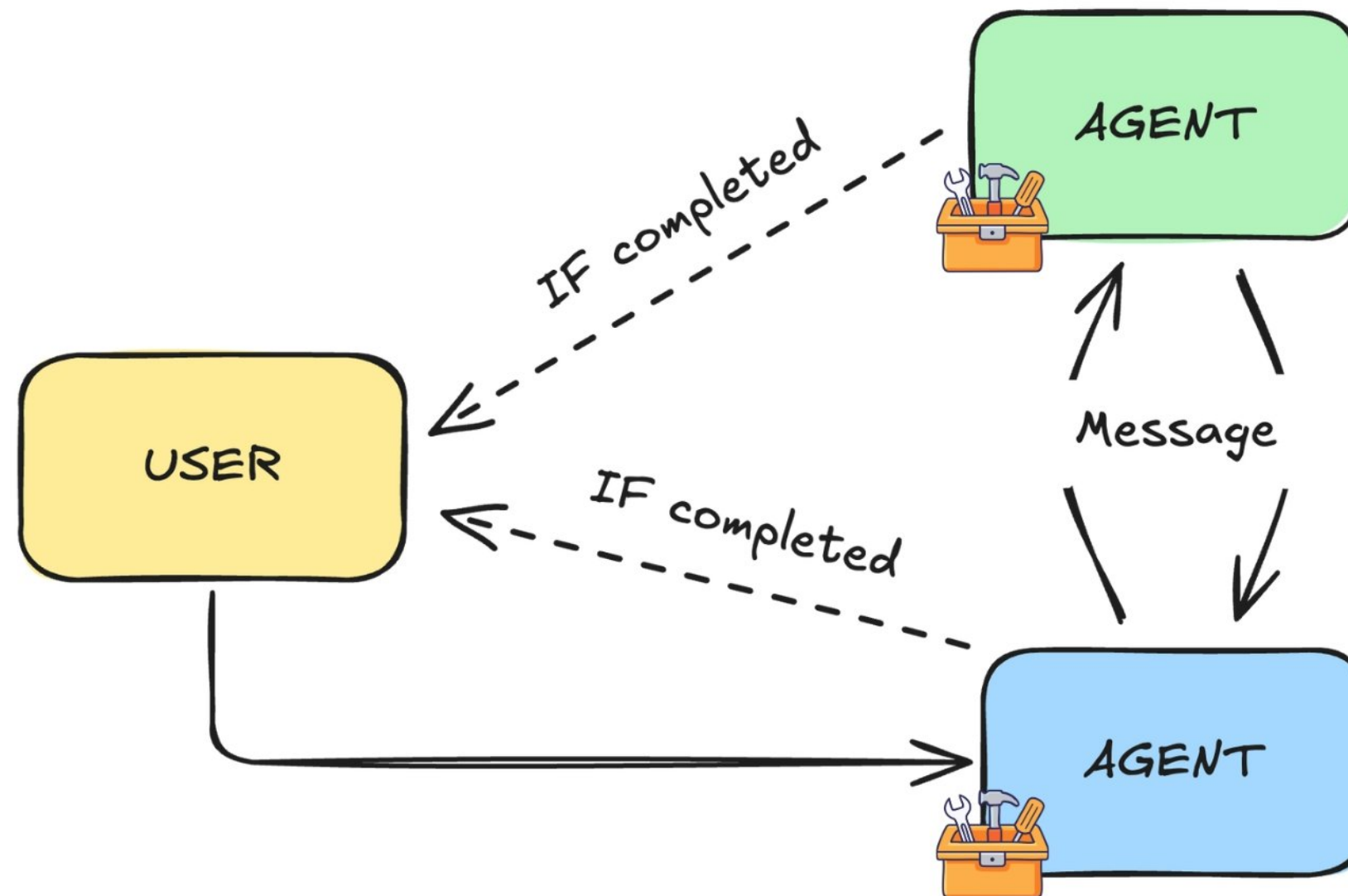
Swarm multi-agents



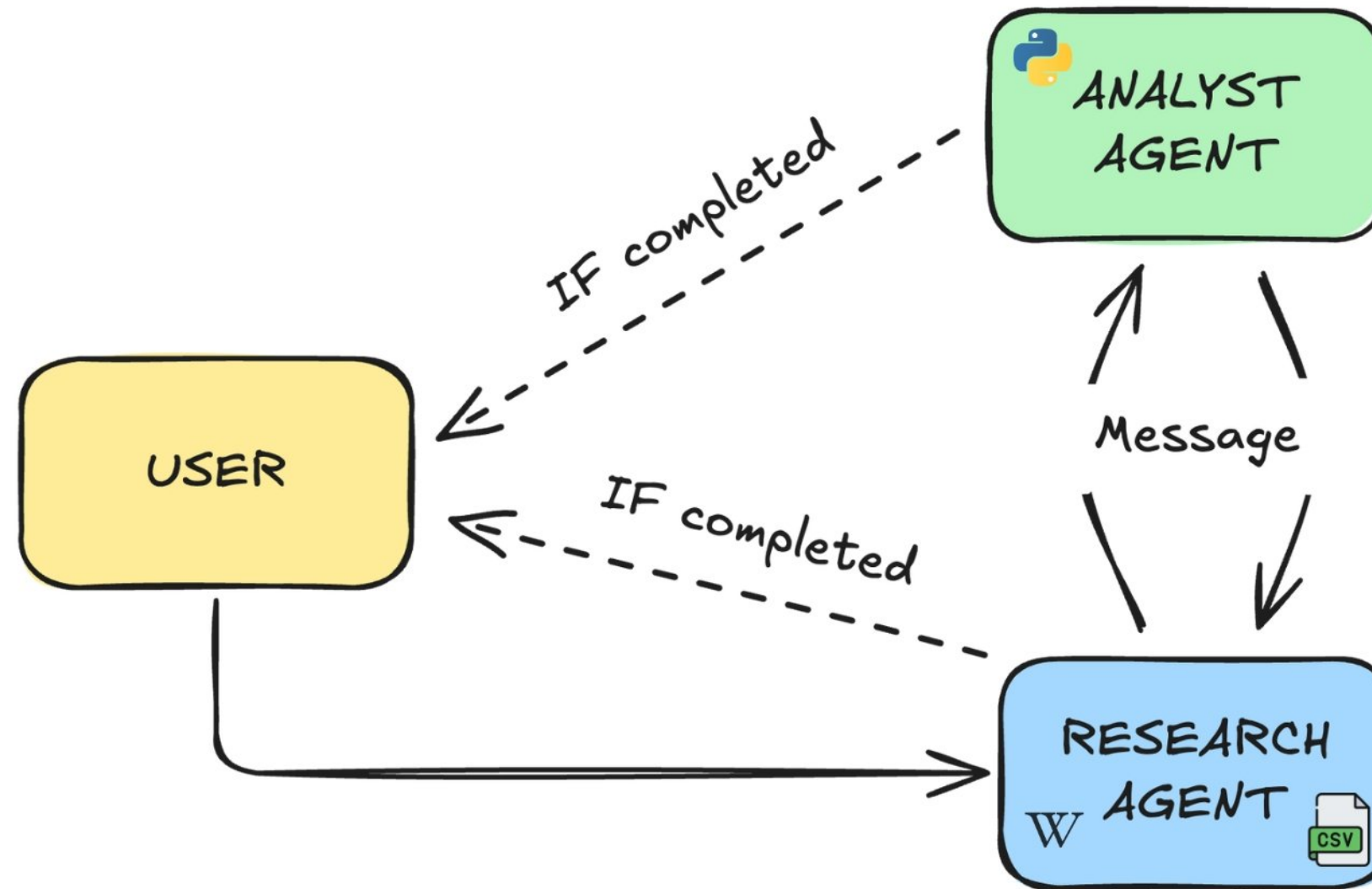
Swarm multi-agents



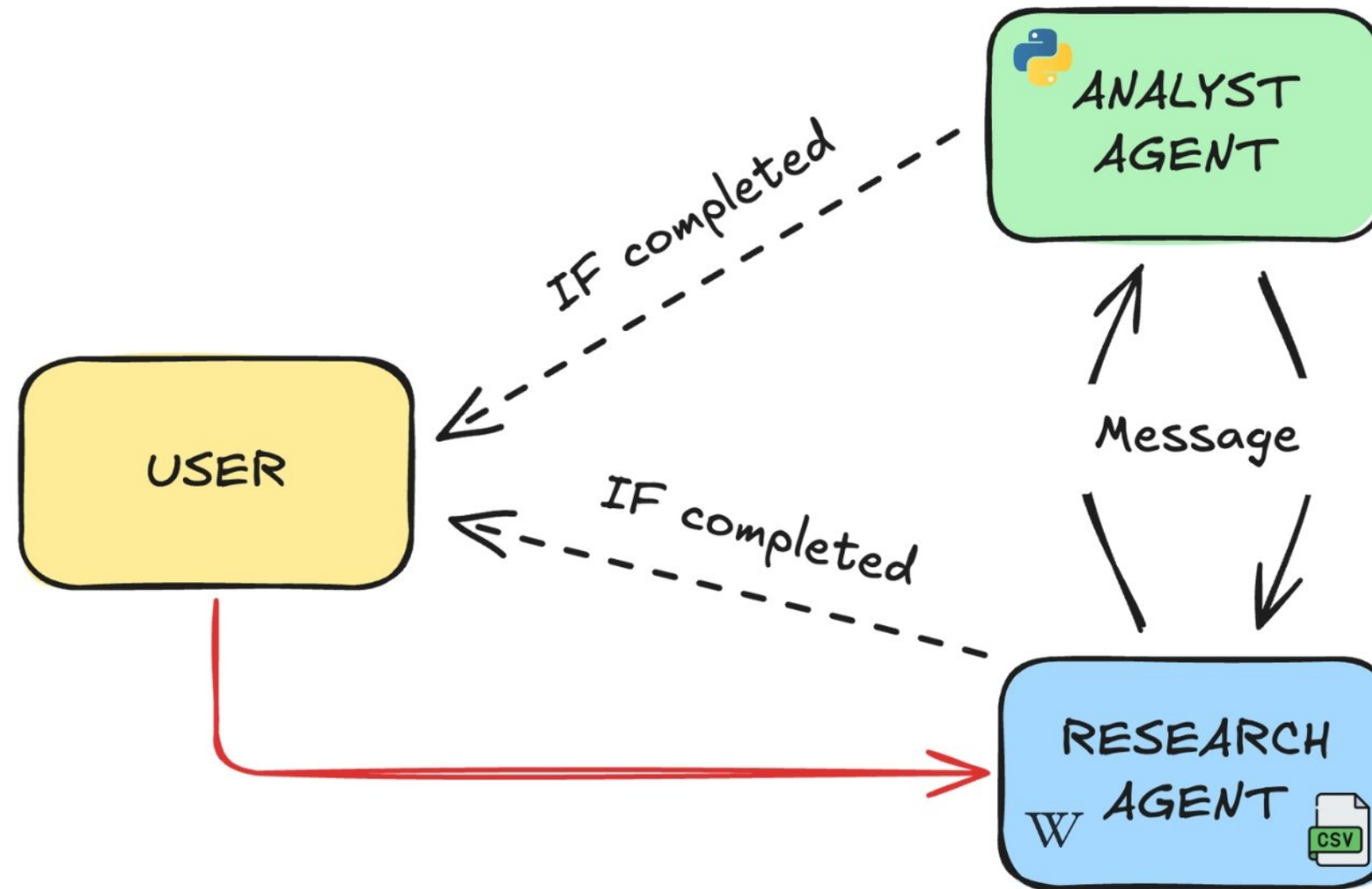
Swarm multi-agents



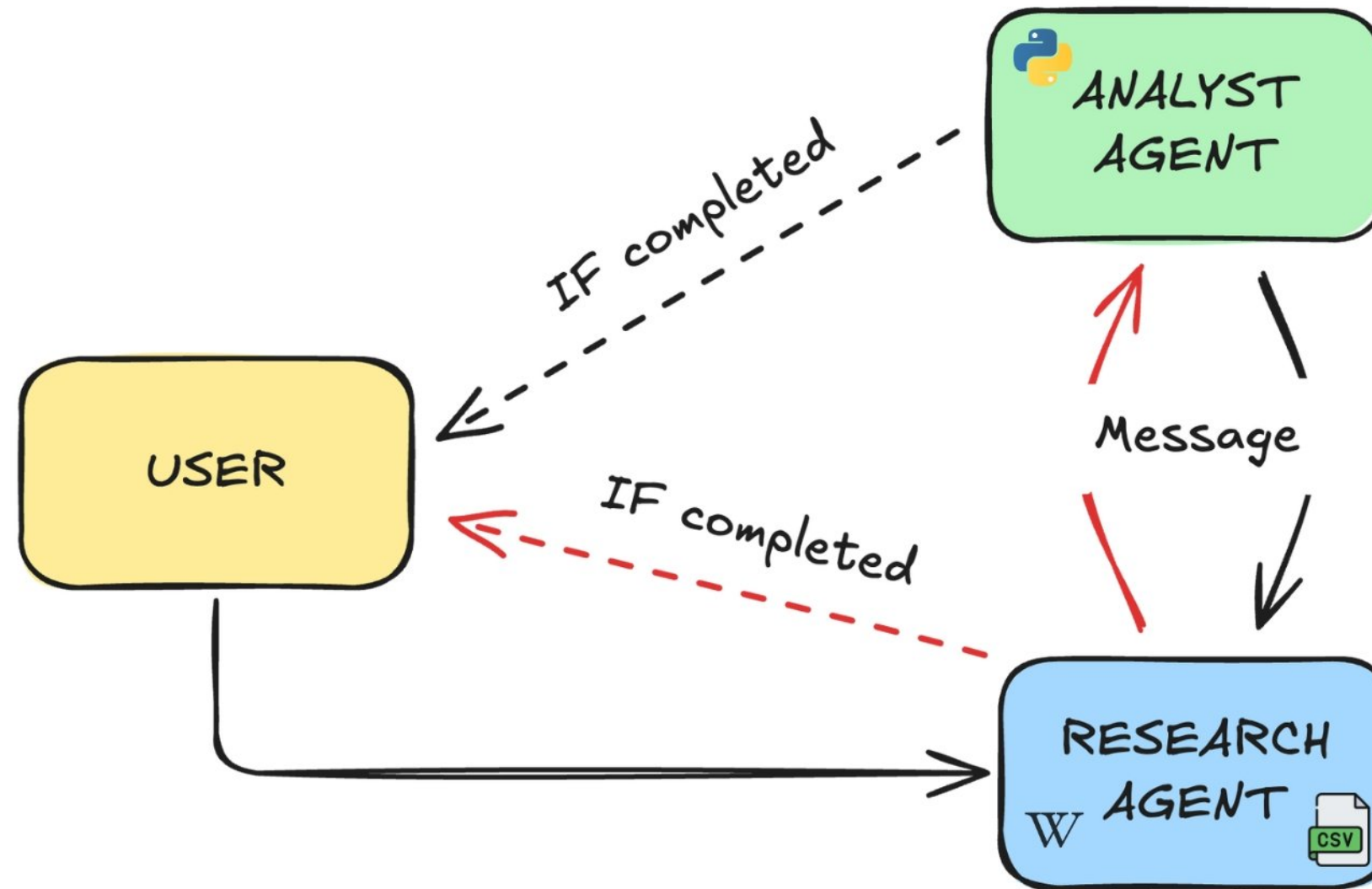
Swarm multi-agents



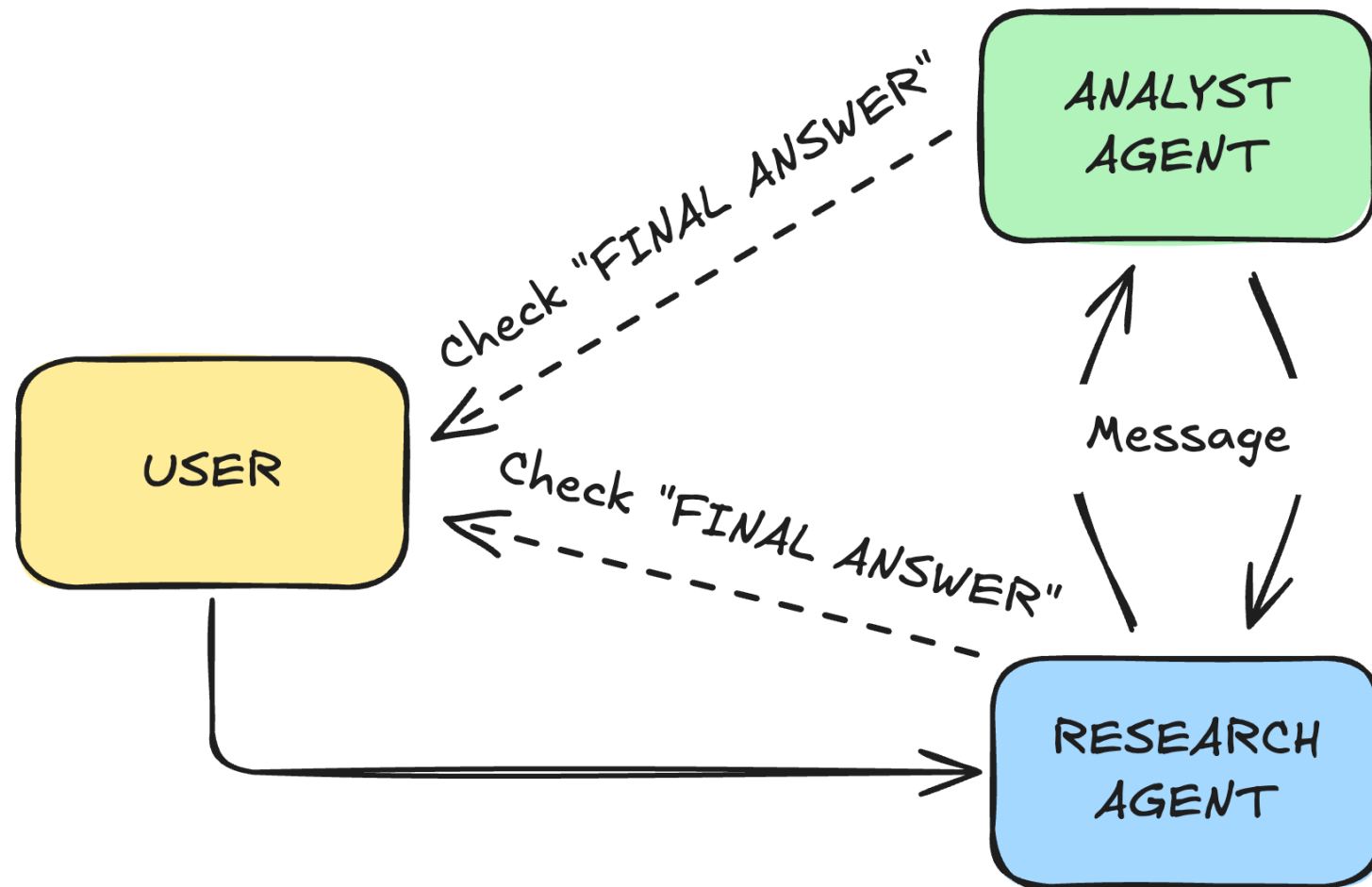
Swarm multi-agents



Swarm multi-agents



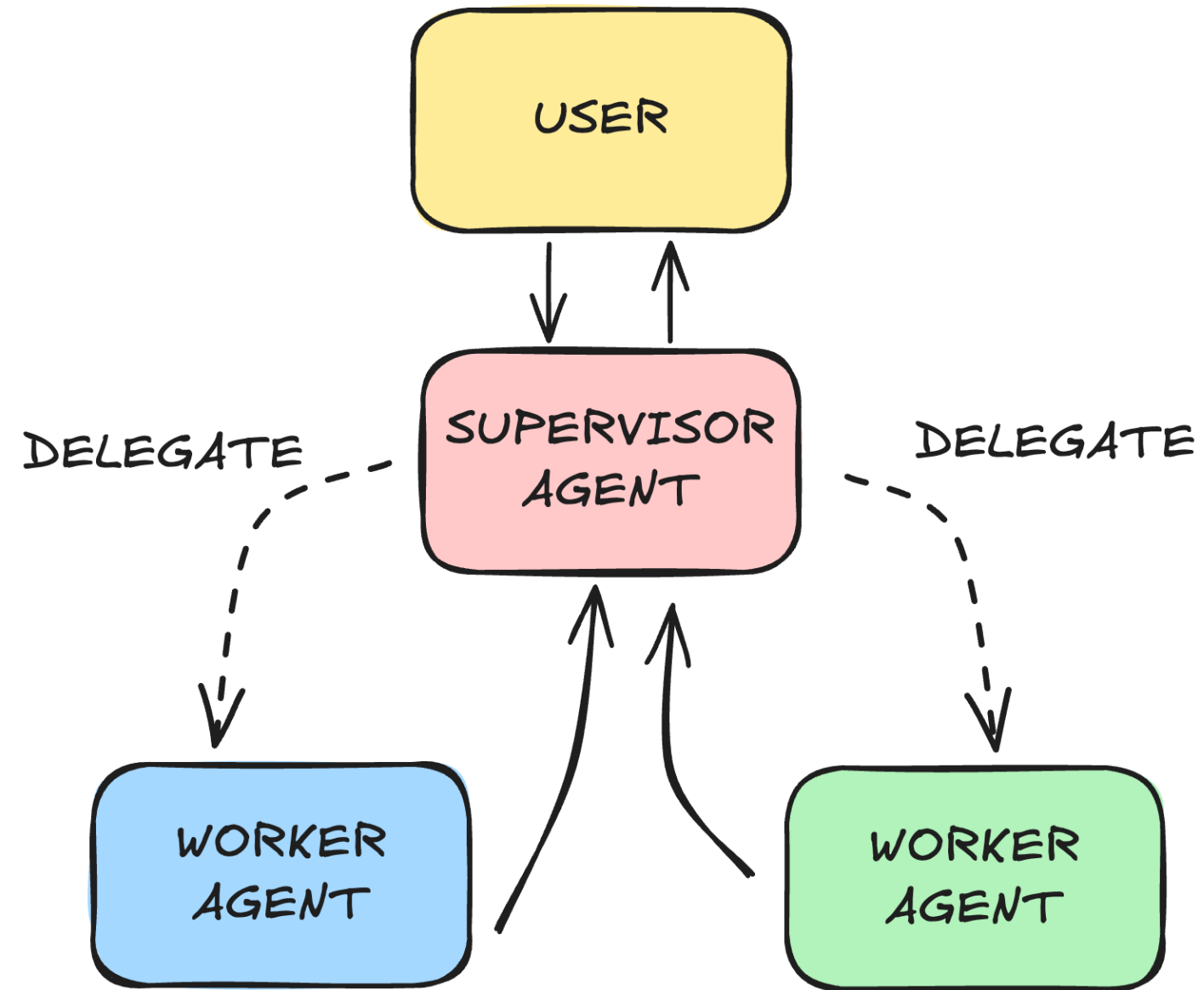
Swarm multi-agents



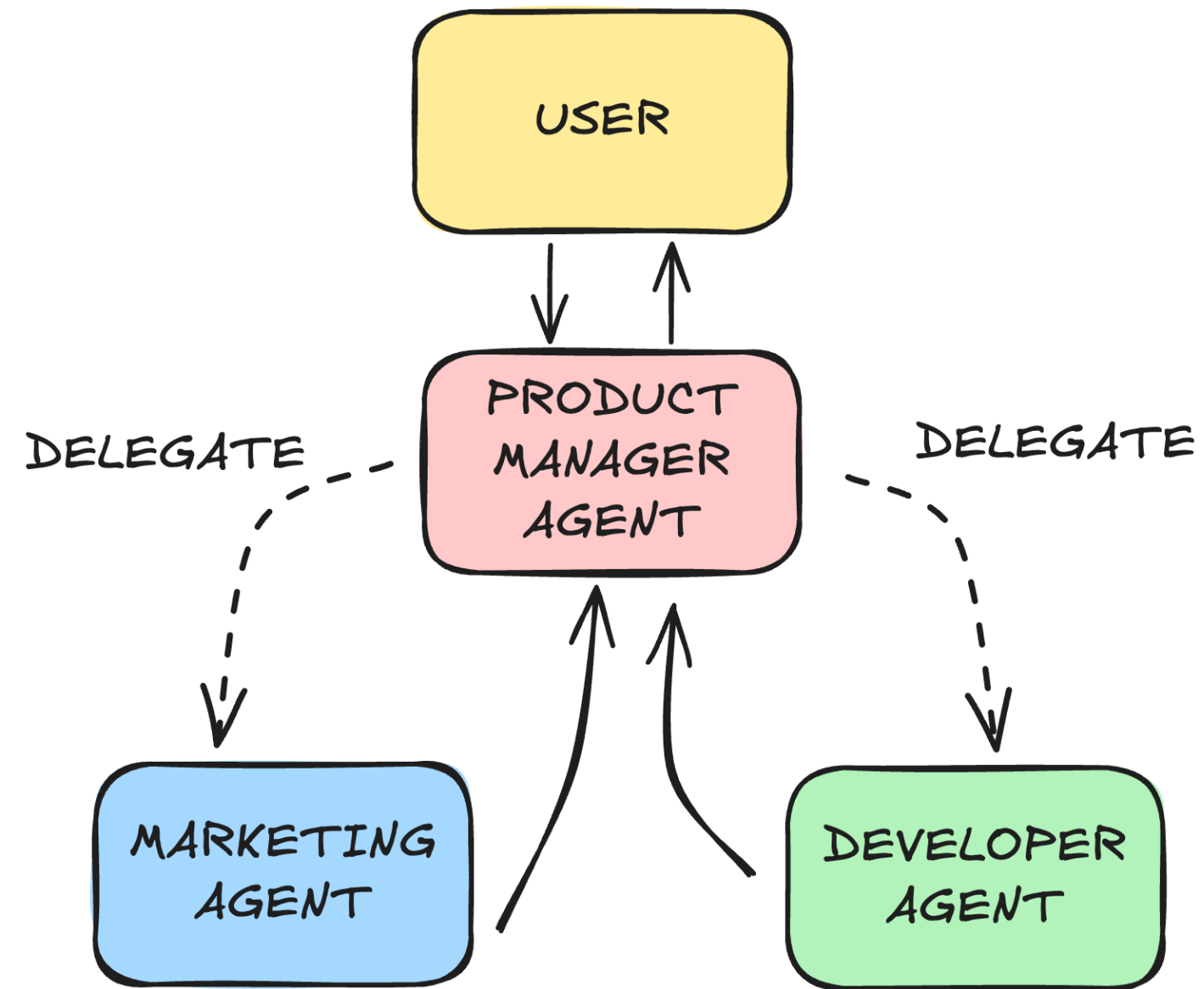
Challenges

1. Agent collaboration
 2. Choosing a starter agent
- Solution → a supervisor!

Supervisor multi-agents



Example: website creator system

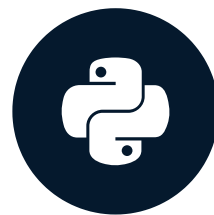


Let's practice!

MULTI-AGENT SYSTEMS WITH LANGGRAPH

Another agent!

MULTI-AGENT SYSTEMS WITH LANGGRAPH



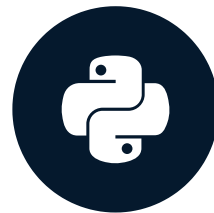
Full Name
Instructor

Let's practice!

MULTI-AGENT SYSTEMS WITH LANGGRAPH

Supervisor agents

MULTI-AGENT SYSTEMS WITH LANGGRAPH



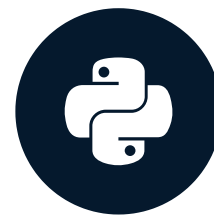
Full Name
Instructor

Let's practice!

MULTI-AGENT SYSTEMS WITH LANGGRAPH

Wrap-up

MULTI-AGENT SYSTEMS WITH LANGGRAPH



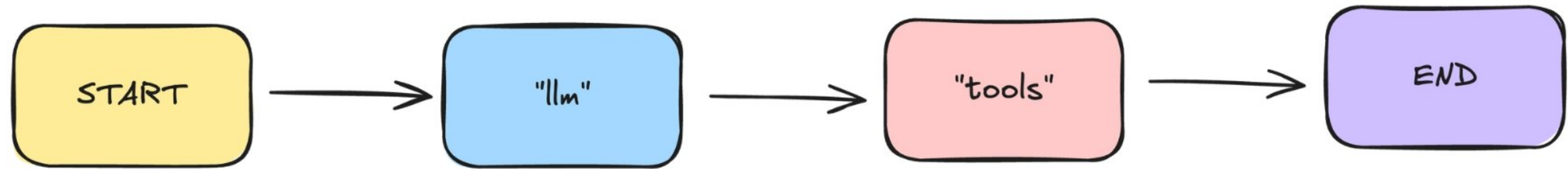
James Chapman

Curriculum Manager, DataCamp

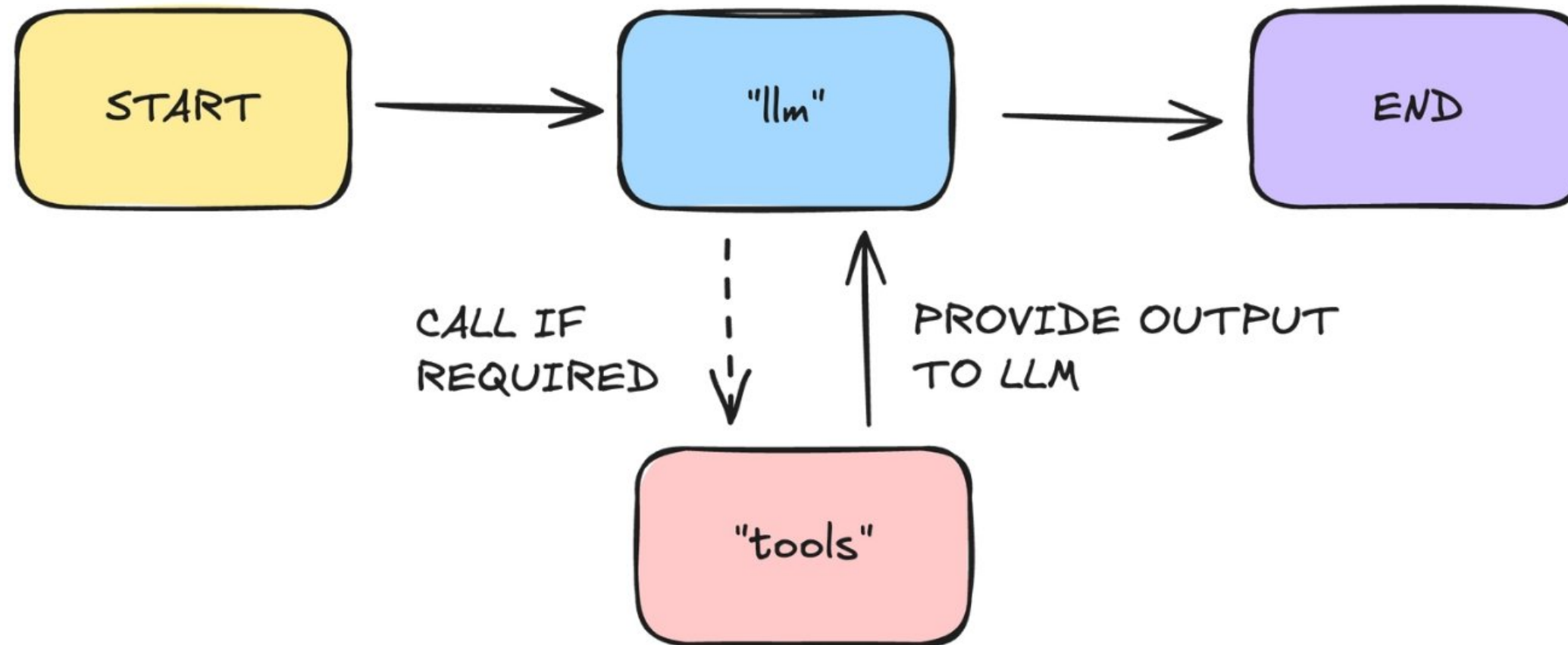
LLM-only graph



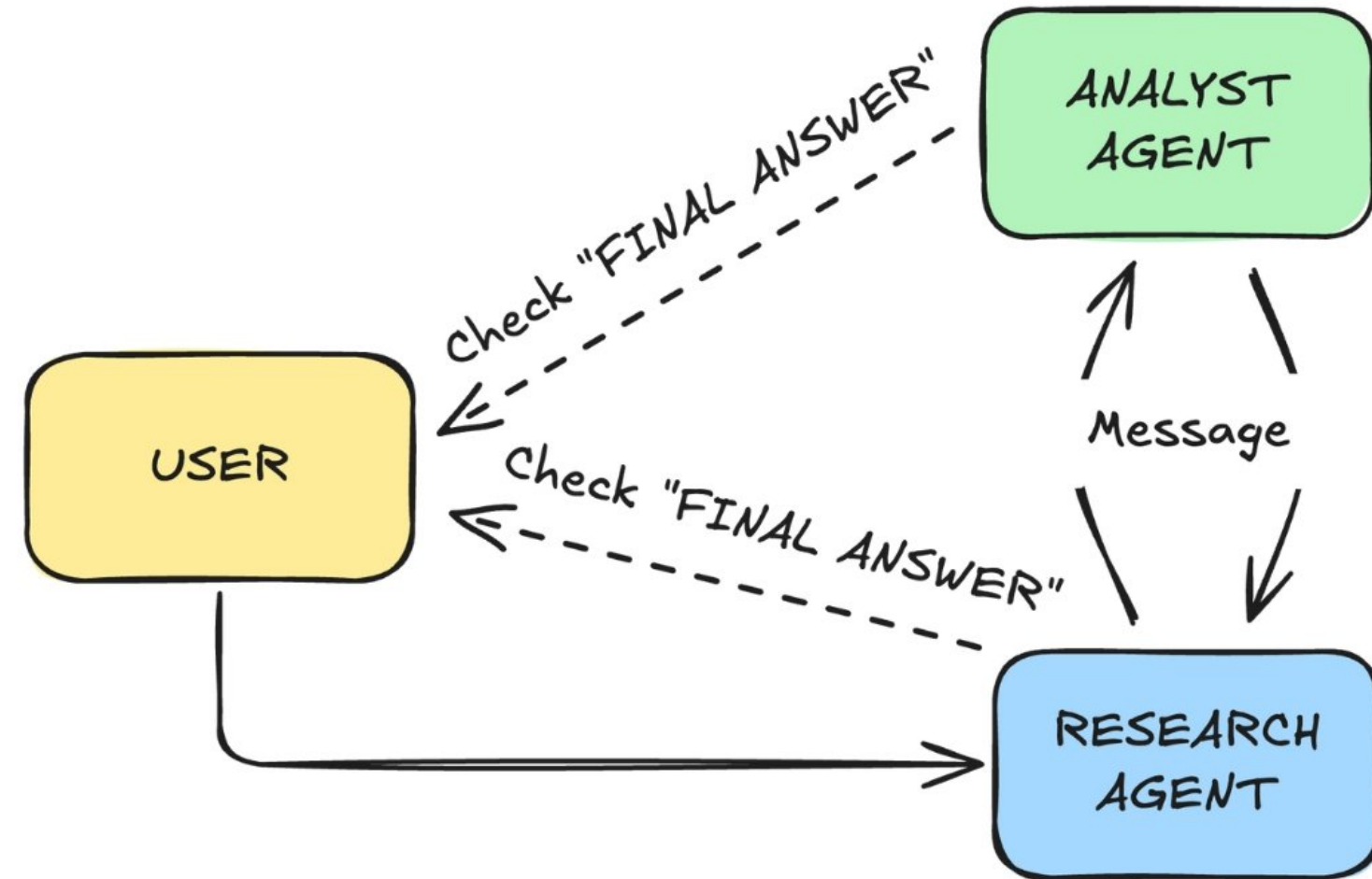
Linear agent



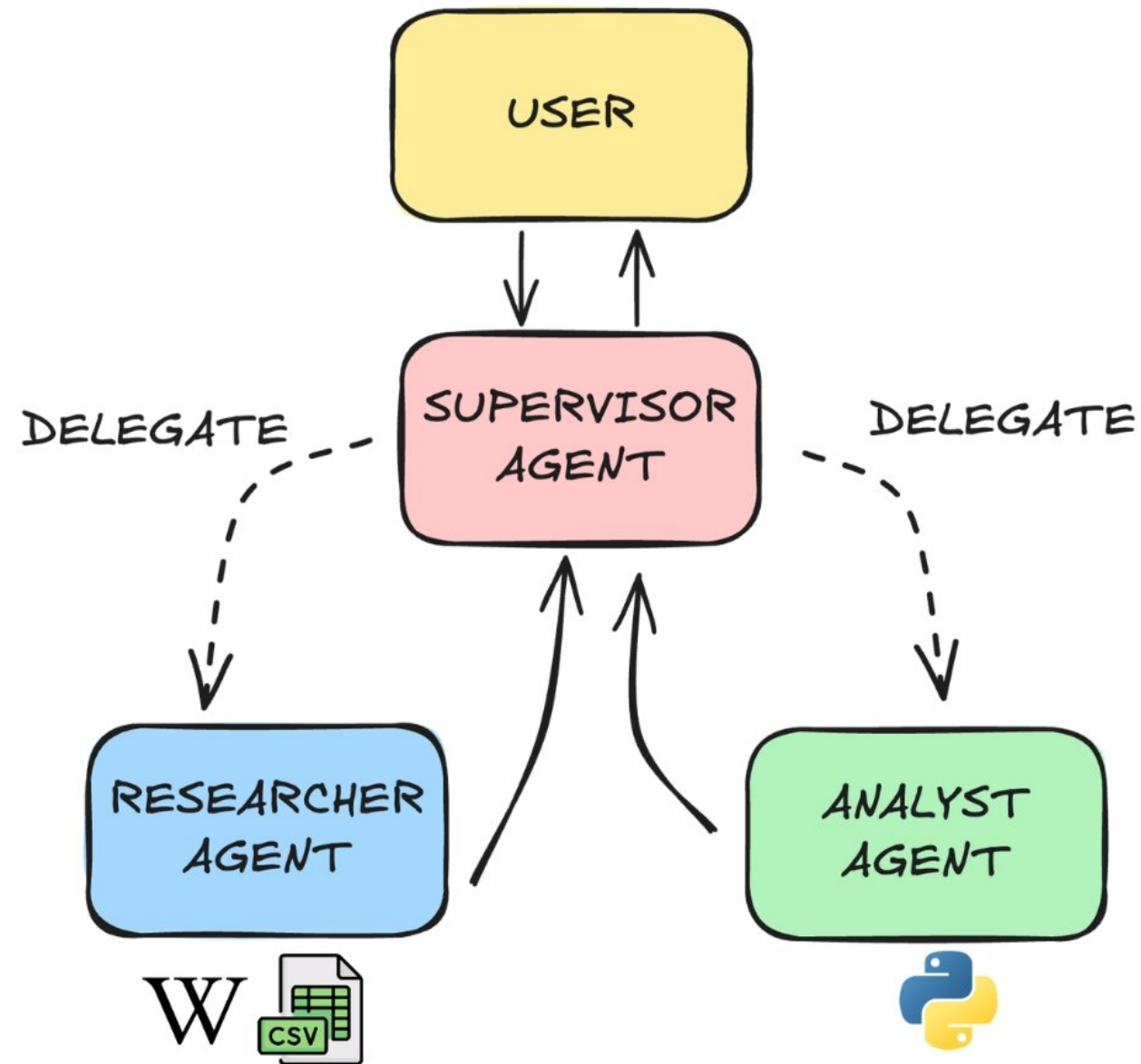
Conditional edges



Swarm multi-agent



Supervisor multi-agent



Let's practice!

MULTI-AGENT SYSTEMS WITH LANGGRAPH